Assignments for the Dynamical Systems Course

May 7, 2026

1 Assignments set I

1.1

Compute characteristic polynomials and eigenvalues for

$$A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix},\ B = \begin{pmatrix} cos(\alpha) & sin(\alpha) \\ -sin(\alpha) & cos(\alpha) \end{pmatrix},\ C = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}$$

1.2

Consider a damped harmonic oscillator:

$$m\ddot{x} + b\dot{x} + kx = 0$$

with $m > 0, b \geq 0, k > 0$.

a) Rewrite as a system of first order differential equations

b) Classify the type of linear system depending on the size of $b \geq 0$

1.3

Consider $\dot{x} = Ax$ with

$$A = \begin{pmatrix} 6 & 6 & 6 \\ 5 & 11 & -1 \\ 1 & -5 & 7 \end{pmatrix}$$

This matrix has an eigenvalue $\lambda = 0$ - see how to show this. What does it tell about the solutions of the system? How many fixed points this system has?

1.4

Find general real solutions for the following systems. Show that the solutions are linearly independent. Draw the phase portrait.

$$\dot{x} = \begin{pmatrix} 6 & -3 \\ 2 & 1 \end{pmatrix} x$$

$$\dot{x} = \begin{pmatrix} -2 & 1 \\ -4 & 3 \end{pmatrix} x$$

$$\dot{x} = \begin{pmatrix} -4 & 1 \\ -1 & -2 \end{pmatrix} x$$

$$\dot{x} = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix} x$$

$$\dot{x} = \begin{pmatrix} 5 & -5 \\ 2 & -1 \end{pmatrix} x$$

1.5

For a non-degenerate matrix A (invertable, non-zero determinant), show that it has the same eigenvectors as its inverse A^{-1}. What is the relation between the eigenvalues of these matrices?

1.6

Are these three vectors linear independent?

$$u = \begin{pmatrix} 1 \\ 3 \\ 2 \\ 1 \end{pmatrix} \quad v = \begin{pmatrix} 4 \\ 9 \\ 5 \\ 4 \end{pmatrix} \quad w = \begin{pmatrix} 3 \\ 7 \\ 4 \\ 3 \end{pmatrix}$$

1.7

$A = \begin{pmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & -1 & -1 \\ 2 & 4 & 0 & -1 \\ -1 & -2 & 1 & 2 \end{pmatrix}$. Find eigenvalues and eigenvectors.